Computer Networks
CL3001

LAB - 08
Wireless Network, NAT

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES,
KARACHI CAMPUS
FAST SCHOOL OF COMPUTING (AI & DS, CS, CY, SE)
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Computer Networks Lab 08

Course: Computer Networks (CL3001) **Semester:** Fall 2025
Lab Prepared By: Mr. Sameer Faisal

Note:
- Maintain discipline during the lab.
- Listen and follow the instructions as they are given.
- Just raise hand if you have any problem.
- Completing all tasks of each lab is compulsory.
- Get your lab checked at the end of the session.

Lab Objective

- Introduction to Wireless Network.
- Importance of Wireless Technology.
- Working of Wireless Network.
- Configuration of Wireless Network.
- Introduction to NAT & its Types.
- Configuration of NAT.
- Wireshark of NAT.

Wireless Network

1. Introduction to Wireless Network

A wireless network allows devices to stay connected to the network while roaming without the need for wires. Because access points amplify Wi-Fi signals, a computer may be far away from a router and still be connected to the network. When you connect to a Wi-Fi hotspot at a café or similar public area, you are connecting to that organization's wireless network.

The main difference between a wireless and a wired network is that a wired network requires wires to connect devices, such as laptops or desktop computers, to the Internet or another network. A wired network, as opposed to a wireless network, has numerous disadvantages. The primary disadvantage is that a router is permanently connected to your computer. The most common wired networks employ cables connected to an Ethernet port on the network router and a computer or other equipment on the other end.

2. Importance of Wireless Technology

However, delving into a specific technology at this point is getting ahead of the tale. Regardless of how the protocols are constructed or what sort of data they carry; wireless networks have some important advantages.

The most obvious benefit of wireless networking is mobility. Users of wireless networks can connect to existing networks and then roam freely. Because the phone connects the user via cell towers, a mobile phone user can travel kilometers in a single call.

Initially, mobile telephone was prohibitively expensive. Due to the exorbitant price, it was only used by highly mobile professionals such as sales managers and important executive decision-makers who needed to be accessible at any time and from any location. Mobile telephone, on the other hand, has proven to be a valuable service that is becoming increasingly popular.

Wireless networks often provide a lot of flexibility, which translates to quick installation. Wireless networks link users to an existing network using a variety of base stations.

3. Working of Wireless Network

A wireless local area network (WLAN) connects a collection of computers in the same way that a wired network does. Because "wireless" does not require costly cabling, the major benefit is that it is generally easier, faster, and less expensive to set up.

Building a network by dragging cables over an office's walls and ceilings, on the other hand, may be time-consuming and costly. Even if you currently have a wired network, a wireless network might be a cost-effective method to extend or expand it.

Radio Frequency (RF) technology, a frequency related with radio wave transmission within the electromagnetic spectrum, is used to power wireless networks. When an RF current is sent into an antenna, it creates an electromagnetic field that can travel over space.

A wireless network's core is a mechanism known as an access point (AP). The fundamental function of an access point is to transmit a wireless signal that computers can detect and tune into. Because wireless networks are frequently linked to wired networks, access points frequently serve as a portal to the resources of the wired network, such as an Internet connection

To connect to an access point and join a wireless network, computers must be equipped with wireless network adapters. These are usually incorporated into the device, but if not, any computer or notebook can be turned wireless-capable by connecting an add-on adapter to an empty expansion slot, USB port, or, in the case of notebooks, a PC card slot.

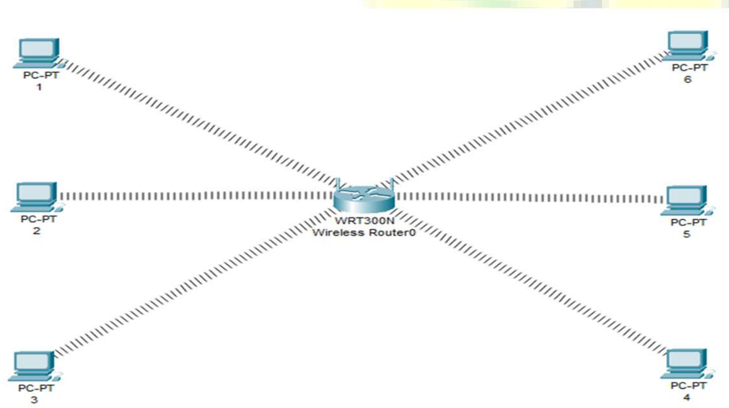


Figure 1

4. Configuration of Wireless Network

The above topology mentioned in figure 1 we have six pc connected with Linksys Wireless routers.

1. DHCP is configured and enabled on Wireless router.
2. IP pool for DHCP is 192.168.0.100 to 192.168.0.150.
3. PC are configured to receive IP from DHCP Server.
4. No security is configured.
5. Default SSID is configured to Default.
6. Topology is working on infrastructure mode.
7. Default user name and password (on router) is admin.
8. IP of wireless router is set to 192.168.0.1.

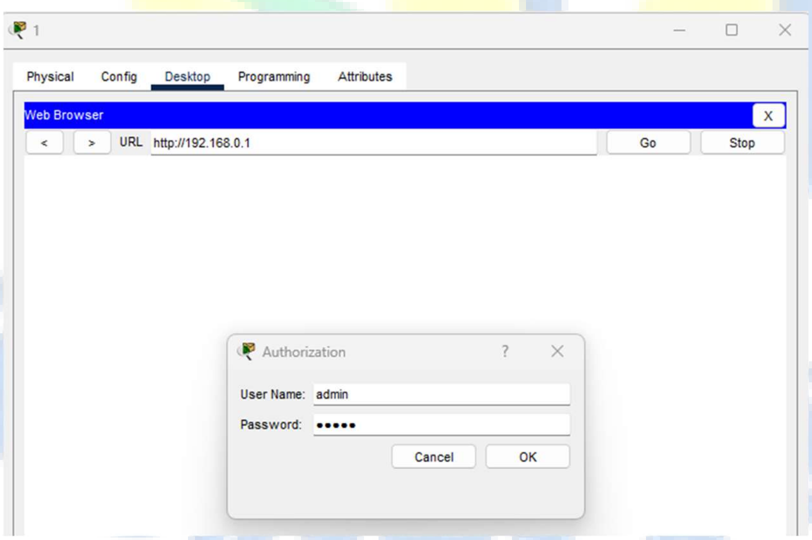
Now we will perform the following tasks on the given above topology i.e., Figure 1:

- Change SSID to any name of your choice.
- Change IP address of router and end devices.
- Configure Static IP on PC and Wireless Router.
- Secure your network by configuring WAP key on Router.
- Connect PC by using WAP key.

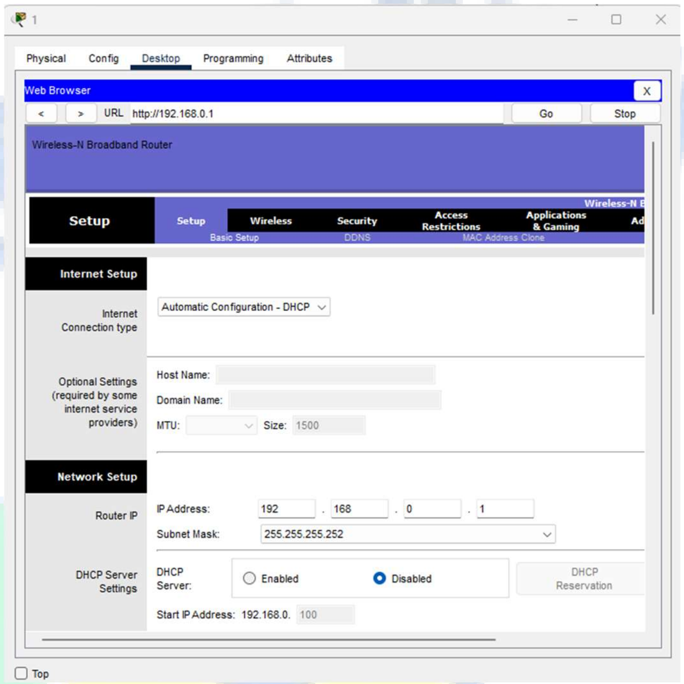
To complete the above tasks, we will follow this step-by-step guide of how to configure wireless network.

As given in question our network is running on 192.168.0.0 network and all PC's are DHCP clients and functioning properly. So, we will first connect to given Wireless router to turn off the DHCP Services.

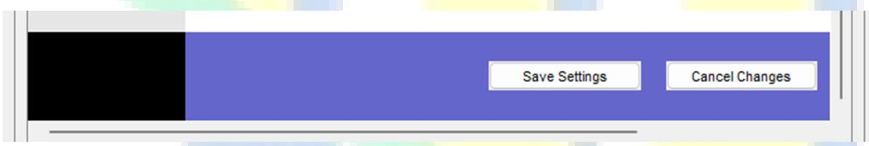
Double click on any PC and select Web Browser. As given, IP of Wireless router is 192.168.0.1 so give it in Web browser and press enter, now it will ask for authentication which is also given above. Default username is admin and password is also admin.

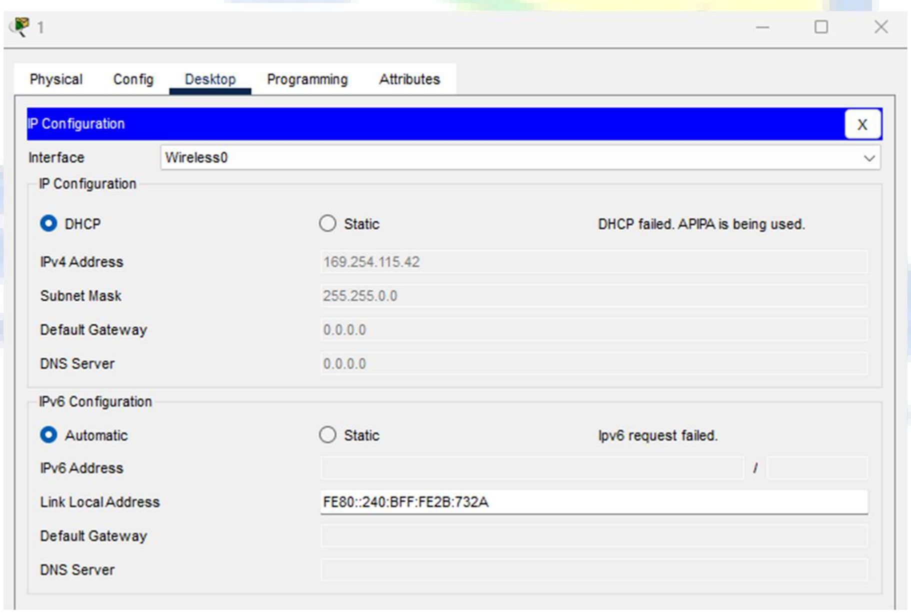


This will bring GUI mode of Wireless router. Scroll down screen to Network Step and Select Disable DHCP.

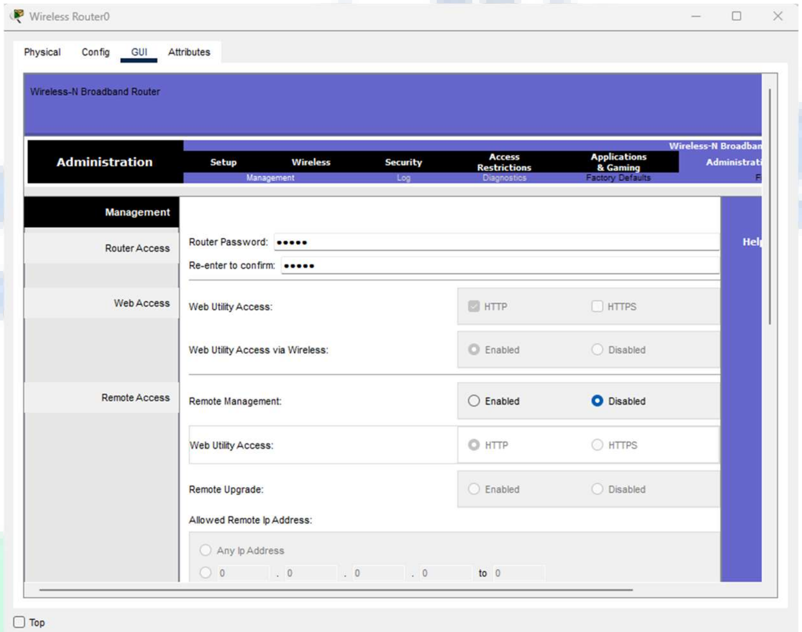


Scroll to the end of the page and click on save setting this will save setting click on continue for further settings.

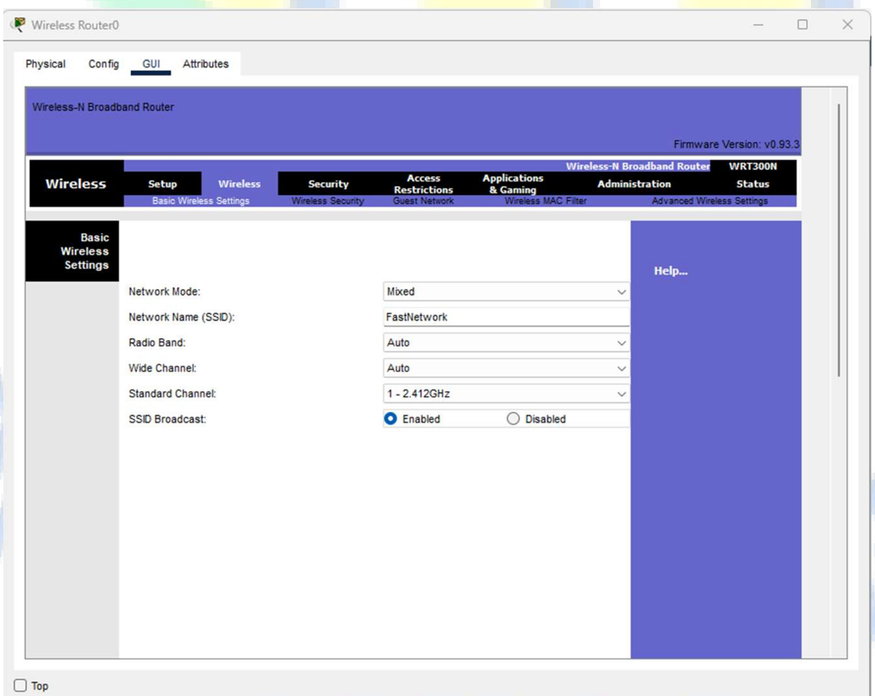




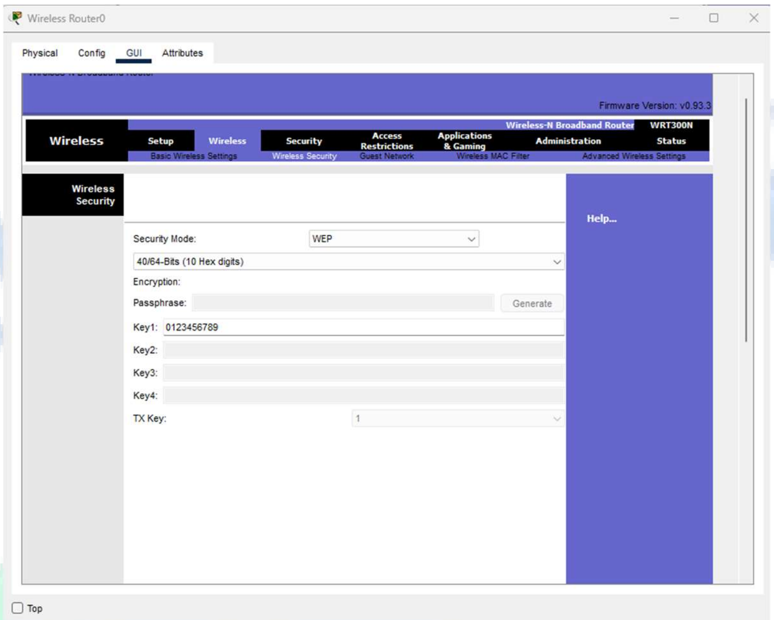
Move to Router directly, select Administration from top Manu and change password to test and go in the end of page and Click on Save Setting.



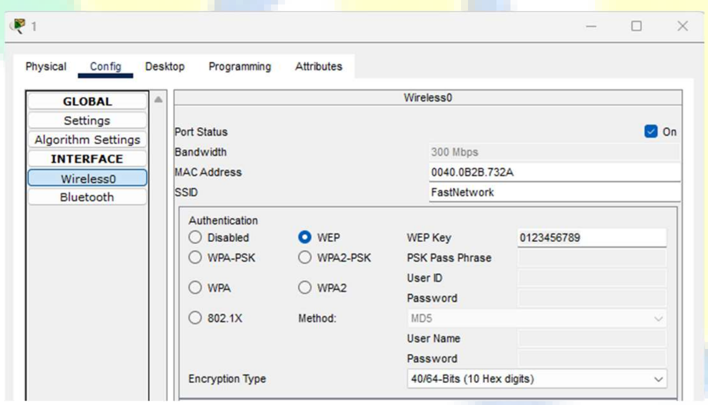
Now click on wireless tab in the router and set default SSID to **FastNetwork** (you can name it something else too).



Now Select wireless security and change Security Mode to WEP:



Again, go at the end of page and Click on Save Setting.
Specify WEP Key on every PC after setting authentication on Router as shown on PC1 below:

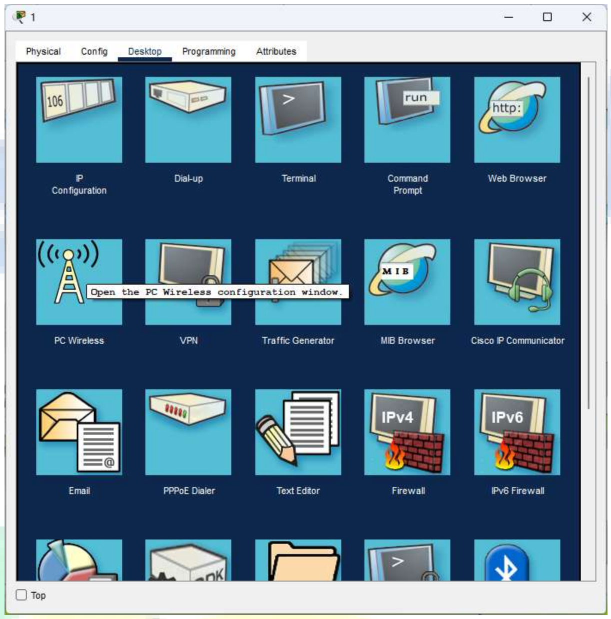


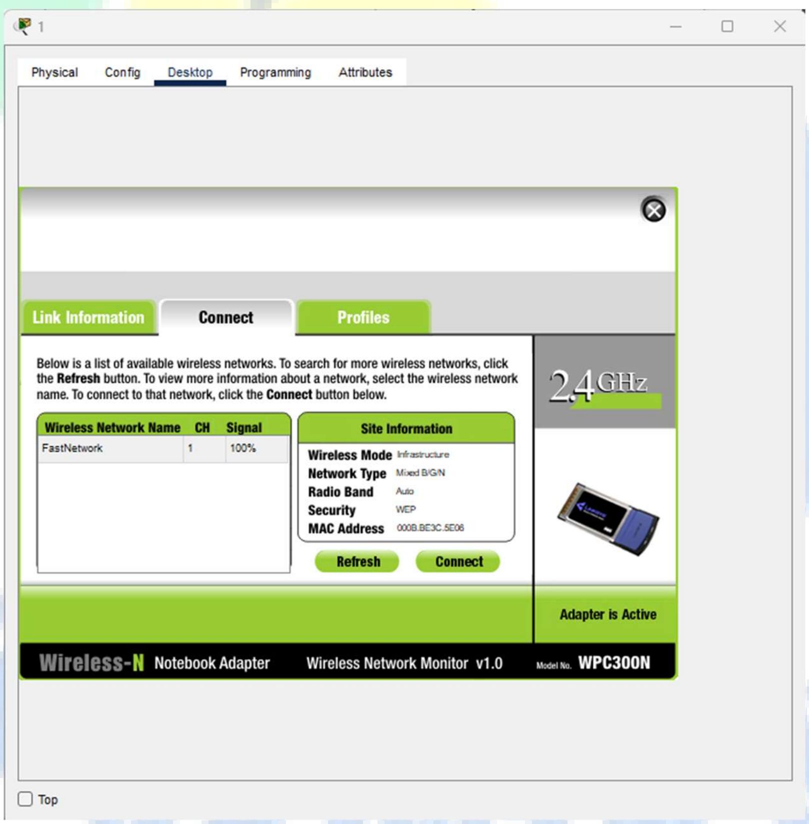
Now we have completed all given task on Wireless router and all PCs are now also given IPs through DHCP. (You can also use static configurations)

PC	IP	Subnet Mask	Default Gateway
1	192.168.0.2	255.255.255.0	192.168.0.1
2	192.168.0.4	255.255.255.0	192.168.0.1
3	192.168.0.5	255.255.255.0	192.168.0.1
4	192.168.0.6	255.255.255.0	192.168.0.1
5	192.168.0.7	255.255.255.0	192.168.0.1
6	192.168.0.8	255.255.255.0	192.168.0.1

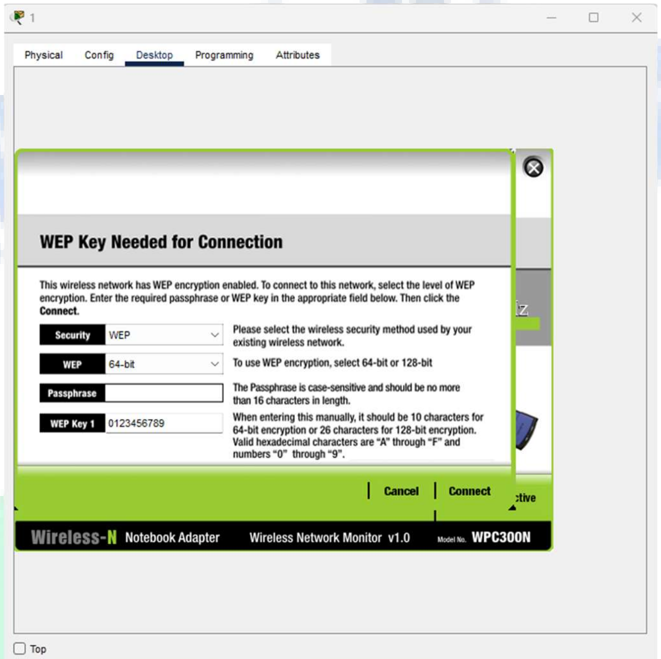
Note: DHCP configurations might differ from the above table.

Now it's time to connect PC's from Wireless router. To do so click PC select Desktop click on PC Wireless:





Click on connect tab and click on Refresh button. It will ask for WEP key insert 0123456789 and click connect.



It will connect you with wireless router as shown below:

NAT

5. Introduction to NAT & Its Types

Network Address Translation (NAT) is a mechanism by which a device modifies a packet's TCP/IP address/port number and maps the IP address from one realm to another (usually from private IP address to public IP address and vice versa). This is accomplished by the NAT device allocating a temporary port number on the public side of the NAT when forwarding outbound packets from the internal host to the Internet, maintaining this mapping for a predetermined period of time, and forwarding inbound packets received from the Internet on this public port back to the internal host.

NAT devices are used to avoid the exhaustion of IPv4 address space by allowing multiple hosts to share a single public/Internet address. Also due to its mapping nature (i.e., a mapping can only be created by a transmission from an internal host), NAT device is preferred to be installed even when IPv4 address exhaustion is not a problem (for example when there is only one host at home), to provide some sort of security/shield for the internal hosts against threats from the Internet.

Despite the fact that NAT provides some shields for the internal network, one must distinguish NAT solution from firewall solution. NAT is not a firewall solution. A firewall is a security solution designed to enforce the security policy of an organization, while NAT is a connectivity solution to allow multiple hosts to use a single public IP address. Understandably both functionalities are difficult to separate at times, since many (typically consumer) products claims to do both with the same device and simply label the device a “NAT box”. But we do want to make this distinction rather clear, as PJNATH is a NAT traversal helper and not a firewall bypass solution (yet).

Following are the types of NAT:

1. **Static NAT (Network Address Translation):** Static NAT (Network Address Translation) is one-to-one mapping of a private IP address to a public IP address. Static NAT (Network Address Translation) is useful when a network device inside a private network needs to be accessible from internet.

2. **Dynamic NAT (Network Address Translation):** Dynamic NAT can be defined as mapping of a private IP address to a public IP address from a group of public IP addresses called as NAT pool. Dynamic NAT establishes a one-to-one mapping between a private IP address to a public IP address. Here the public IP address is taken from the pool of IP addresses configured on the end NAT router. The public to private mapping may vary based on the available public IP address in NAT pool.

3. **PAT (Port Address Translation):** Port Address Translation (PAT) is another type of dynamic NAT which can map multiple private IP addresses to a single public IP address by using a technology known as Port Address Translation.

6. Configuration of NAT

6.1.Static NAT Configuration

Static NAT is used to do a one-to-one mapping between an inside address and an outside address. Static NAT also allows connections from an outside host to an inside host. Usually, static NAT is used for servers inside your network. For example, you may have a web server with the inside IP address 192.168.0.10 and you want it to be accessible when a remote host makes a request to 209.165.200.10. For this to work, you must do a static NAT mapping between those to IPs. In this example, we will use the Fast Ethernet 0/1 as the inside NAT interface, the interface connecting to our network, and the Serial 0/0/0 interface as the outside NAT interface, the one connecting to our service provider.

Working Steps on Router:

Router(config)#ip nat inside source static 192.168.0.10 209.165.200.10
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip nat inside
Router(config-if)#interface Serial 0/0/0
Router(config-if)#ip nat outside

6.2.Dynamic NAT Configuration

Dynamic NAT is used when you have a “pool” of public IP addresses that you want to assign to your internal hosts dynamically. Don’t use dynamic NAT for servers or other devices that need to be accessible from the Internet. In this example, we will define our internal network as 192.168.0.0/24. We also have the pool of public IP addresses from 209.165.200.226 to 209.165.200.240 and our assigned netmask is 255.255.255.224. When you configure dynamic NAT, you have to define an ACL to permit only those addresses that are allowed to be translated.

Working Steps on Router:

Router(config)#ip nat pool NAT-POOL 209.165.200.226 209.165.200.240 netmask 255.255.255.224
Router(config)#access-list 1 permit 192.168.0.0 0.255.255.255
Router(config)#ip nat inside source list 1 pool NAT-POOL
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip nat inside Router
(config-if)#interface Serial 0/0/0
Router(config-if)#ip nat outside

7. NAT Overload (PAT) Configuration

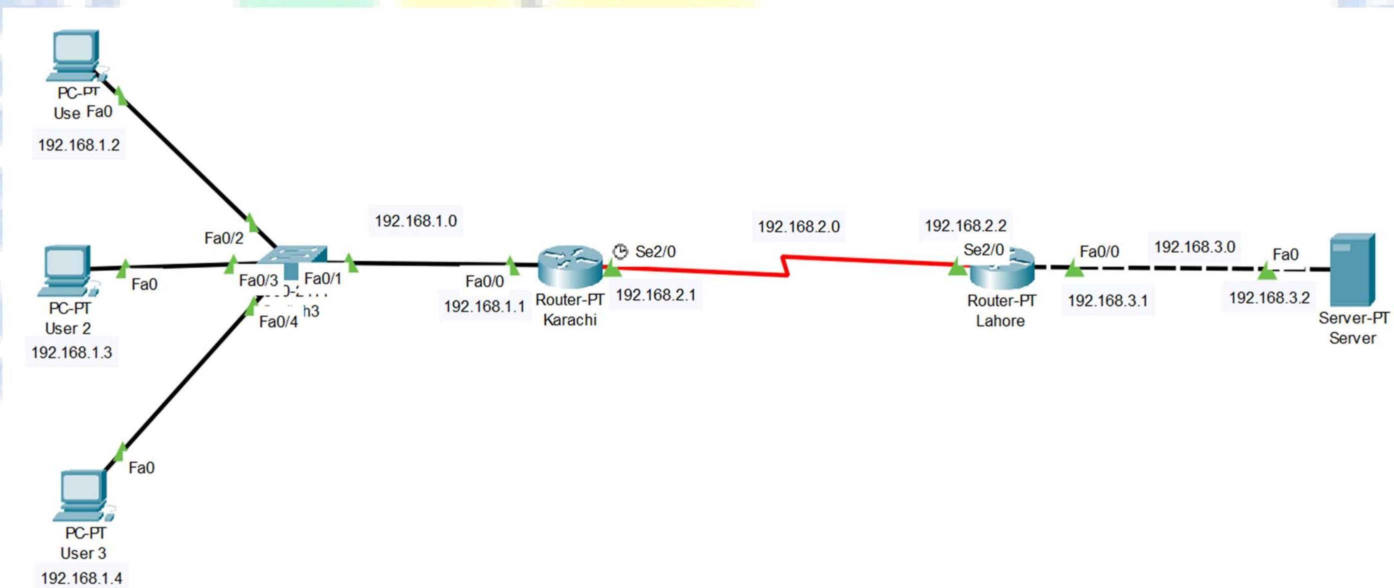
NAT Overload, sometimes also called PAT, is probably the most used type of NAT. You can configure NAT overload in two ways, depending on how many public IP address you have available. The first case, and one of the most often seen cases, is that you have only one public IP address allocated by your ISP. In this case, you map all your inside hosts to the available IP address. The configuration is almost the same as for dynamic NAT, but this time you specify the outside interface instead of a NAT pool.

Working Steps on Router:

Router(config)#access list 1 permit 192.168.0.0 0.255.255.255
Router(config)#ip nat inside source list 1 interface serial 0/0/0 overload
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip nat inside
Router(config-if)#interface Serial 0/0/0
Router(config-if)#ip nat outside

Example

In this example we configure NAT overload (PAT) on the router which is located in Karachi and static NAT is configured on Lahore router. A simple topology is shown below:



Assigning IP address to Karachi Router:

```
Router>enable
Router# configure terminal
Router(config)#
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface Serial2/0
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#clock rate 64000
R1(config-if)#bandwidth 64
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
```

Assigning IP address to Lahore Router:

```
Router>enable
Router#configure terminal
Router(config)#hostname R2
R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface Serial2/0
R2(config-if)#ip address 192.168.2.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
```

Configuring NAT Overload (PAT) on router Karachi:

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 1 permit 192.168.1.2 0 0.0.0.0
R1(config)#access-list 1 permit 192.168.1.3 0.0.0.0
R1(config)#access-list 1 deny any
R1(config)#ip nat pool ccna 50.0.0.1 50.0.0.1 netmask 255.0.0.0
R1(config)#ip nat inside source list 1 pool ccna overload
R1(config)#interface FastEthernet 0/0
R1(config-if)#ip nat inside
R1(config-if)#exit
R1(config)#interface Serial 0/0/0
R1(config-if)#ip nat outside
R1(config-if)#exit

Configuring Static NAT on router Lahore:

R2>enable
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip nat inside source static 192.168.3.1 200.0.0.10
R2(config)#interface Serial 2/0
R2(config-if)#ip nat outside
R2(config-if)#exit
R2(config)#interface FastEthernet 0/0
R2(config-if)#ip nat inside
R2(config-if)#exit
R2(config)#

Configuring static route on routers:

R1(config)#ip route 200.0.0.0 255.255.255.0 100.0.0.2
R2(config)#ip route 50.0.0.0 255.0.0.0 100.0.0.1

Now ping server from a PC or open website through web browser using 200.0.0.10 IP address. To check the connectivity after configuring NAT. Use show "ip nat translation command" to verify NAT implementation on router.

```
R1#show ip nat translation
Pro  Inside global      Inside local       Outside local      Outside global
icmp 50.0.0.1:1         10.0.0.20:1        200.0.0.10:1       200.0.0.10:1
icmp 50.0.0.1:2         10.0.0.20:2        200.0.0.10:2       200.0.0.10:2
icmp 50.0.0.1:3         10.0.0.20:3        200.0.0.10:3       200.0.0.10:3
icmp 50.0.0.1:4         10.0.0.20:4        200.0.0.10:4       200.0.0.10:4
tcp 50.0.0.1:1024       10.0.0.10:1025     200.0.0.10:80      200.0.0.10:80
tcp 50.0.0.1:1025       10.0.0.20:1025     200.0.0.10:80      200.0.0.10:80
```